Mihai-Cristian **Farcaș**

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EDUCATION

- **Babeș-Bolyai University** — **Cluj-Napoca, Romania**
  MSc in Cloud, Network Infrastructure and High-Performance Computation — *2026 - Present*
- **Babeș-Bolyai University** — **Cluj-Napoca, Romania**
  BSc in Computer Science (English), Faculty of Mathematics and Computer Science — *Oct 2023 - June 2026*
  - **Thesis**: Hybrid post-quantum key exchange for WireGuard, the first FIPS 203-compliant hybrid WireGuard implementation on a production Rust codebase.

EXPERIENCE

- **BluePlan** — **Software Engineer (Freelance)**
  Atlas: RAG Platform — *2026 - Present*
  - Built and shipped Atlas end to end: retrieval-augmented chat over technical documents with hybrid dense, sparse and image retrieval fused with RRF, reranking and verbatim citations pinned to the source PDF. Multi-tenant, multilingual and in daily production use. Python, FastAPI, Qdrant, React.
- **Baseline Studios** — **Full-Stack Developer**
  Client Applications — *2026*
  - Delivered full-stack features across multiple client applications (stratosphere.vip, pursuitofx.org, kabu.xyz, esoftplus-landing.vercel.app) using TypeScript and React.
  - **Technologies**: TypeScript, React.
- **GeoDraw** — **Software Developer**
  AI-Powered Design Canvas — *2025*
  - Developed an AI-powered canvas for sketching and wireframing apps with React, TLDraw and LLM APIs, turning natural language into accurate real-time design output on the canvas.
  - **Technologies**: React, TLDraw, LLM APIs.
- **Tvarita** — **Software Development Intern**
  Website Redesign — *April - June 2025*
  - Rebuilt and redesigned a server monitoring site for an optician clinic chain using React and Tailwind CSS, improving the user experience and significantly reducing load times.
  - **Technologies**: React, Tailwind CSS.

PERSONAL PROJECTS

- **pq-boringtun** — **2026**
  - Hybrid post-quantum key exchange for WireGuard: ML-KEM-768 alongside X25519 in the Noise IK handshake, built in Rust on Cloudflare's BoringTun and benchmarked on Apple M4 and Raspberry Pi 5.
  - **Technologies**: Rust, ML-KEM-768, X25519, Noise Protocol, WireGuard.
- **Supernova** — **2026**
  - Live-wallpaper engine for macOS in Swift 6: renders looping video, Shadertoy-style GLSL shaders transpiled to Metal and sandboxed web bundles behind desktop icons, with audio, cursor and motion reactivity, per-display assignment and smart playlists.
  - **Technologies**: Swift 6, Metal, Core Audio, GLSL.
- **mihaicristian.dev** — **2026**
  - Bilingual (EN/RO) personal portfolio site built with Astro: project showcases, long-form writing and a performance-focused SPA-style browsing experience.
  - **Technologies**: Astro, TypeScript.
- **eBPF Network Analysis System** — **2025**
  - Network traffic classification for QUIC protocols, inspecting encrypted flows at the kernel boundary using eBPF

with C, Rust and Python.
  - **Technologies**: C, Rust, Python, eBPF.

- **nvim-search-and-replace** **2025**
  - Project-wide search and replace plugin for Neovim written in Lua: async streaming ripgrep search, live before/after preview, selective replacement and its own undo/redo stack.
  - **Technologies**: Lua, Neovim, ripgrep.

TECHNICAL SKILLS

Programming Languages: Rust, Go, Python, Swift, C/C++, Java, TypeScript/JavaScript, Lua, Shell.

Web Development: React, Next.js, Astro, Tailwind CSS, TLDraw, Vercel, Cloudflare Workers.

Backend, Databases & ML: FastAPI, PostgreSQL, Qdrant, FAISS, LangChain, PyTorch, LLM APIs.

Systems, Networking & Security: eBPF, WireGuard, Noise Protocol, ML-KEM, Metal, Core Audio, GLSL shaders.

Tools: Git, Neovim, ripgrep, Linux, macOS.